# Piter Z. Garcia Bautista

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## Research Summary

Graduate researcher and NSF Robert Noyce Scholar working at the intersection of quantum computing, compiler-level optimization, and reproducible experimentation. Current research spans quantum network optimization, fault-tolerant routing, and fairness-aware decision workflows. Specifically interested in compiler-assisted precision and resource management for fault-tolerant quantum workloads, as explored in the SPQR project and TAFFO-inspired frameworks for quantum error correction code selection.

## Education

**Rochester Institute of Technology** — Expected Aug 2026
*Master of Science, Data Science — GPA 3.9* — *Rochester, NY*

- Concentration: Bioinformatics & Artificial Intelligence.
- Relevant coursework: Foundations of Data Science (DSCI-633), Software Engineering for Data Science (DSCI-644), Applied Data Science I (DSCI-601), Bioinformatics Algorithms (BIOL-630), Data-Driven Knowledge Discovery (ISTE-780).

**University of Rochester, Warner School of Education** — Expected Aug 2026
*Master of Science track, Teaching Computer Science K–12 — GPA 4.0* — *Rochester, NY*

- NSF Robert Noyce Teacher Scholarship Program ($30,000 full funding).
- Advanced Certificate in Leadership in Disability and Inclusive Practices.

**Rochester Institute of Technology** — 2015
*Master of Science, Computer Science* — *Rochester, NY*

- Concentrations: Computer Graphics, Artificial Intelligence, Big Data Analytics.
- Coursework in compiler design, algorithm analysis, and distributed systems.

**Farmingdale State College** — 2012
*Bachelor of Science, Computer Engineering Technology — GPA 3.3* — *Farmingdale, NY*

- Dual degree: Electrical Engineering & Computer Engineering Technology.
- Honors: CSTEP Award & Merit, Dual BS Scholarship.

## Research Experience & Projects

**Rochester Institute of Technology** — Fall 2025 – Present
*Graduate Assistant, Quantum and AI Research* — *Rochester, NY*

- Conduct research on quantum network path optimization and fault-tolerant routing under adversarial and stochastic conditions.

- Develop compiler-level experimental infrastructure for quantum circuit workloads using Qiskit and Cirq.
- Implement and validate quantum error correction code selection workflows in simulation (Stim-compatible pipelines).
- Apply LLVM/MLIR IR-level analysis concepts to quantum compilation pipeline studies; reviewed TAFFO precision-tuning methodology for applicability to quantum resource management.

**RIT & University of Rochester** 2024 – Present
*Lead Researcher, EQUITAS Framework for Diagnostic Justice* *Rochester, NY*

- Developed an interdisciplinary framework integrating fairness-aware ML, bioinformatics, and trauma-informed analysis.
- Built reproducible bias-assessment workflows and reported equity gaps across algorithmic variants.
- Coordinated cross-domain collaboration across computing, education, and health research contexts.

**Rochester Institute of Technology** 2024 – Present
*Principal Investigator, Equitable Bioinformatics Project (ISTE-780)* *Rochester, NY*

- Applied systematic fairness analysis to RNA secondary structure prediction pipelines.
- Implemented reproducible evaluation codebase with automated metric reporting and statistical testing.

## Teaching & Mentoring Experience

**University of Rochester, Warner School of Education** Sep 2024 – Present
*CS Teacher Candidate, K–12 Computer Science Teaching Placement* *Rochester, NY*

- Deliver K–12 computer science instruction in public school placement as part of NSF Noyce Scholar program.
- Design and teach CS curriculum aligned with NYS learning standards; mentor students in computational thinking and algorithmic reasoning.

**Varsity Tutors & Independent Practice** 2021 – Present
*STEM Tutor and Academic Mentor* *Remote & In-Person*

- Tutor Computer Science, Data Science, Mathematics, and Statistics for diverse learners.
- Design culturally responsive instruction and mentoring support for underrepresented students in STEM.

**Rochester Institute of Technology** 2013 – 2015
*Faculty Assistant, Graduate Office* *Rochester, NY*

- Supported graduate program operations, student documentation, and faculty coordination.

## Selected Publications & Academic Contributions

- *Quantum Network Path Optimization and Fault-Tolerant Routing* (manuscript under review, 2025).
- *Designing Equitable AI Systems for Diagnostics: An IDAI700 Research Portfolio* (Project Report, 2025).
- *Equitable Bioinformatics: Enhancing Diagnostic Decision-Making through RNA and Biomarker Data* (ISTE-780 Project Report, 2025).

- *Trauma-Informed Bioinformatics for Diagnostic Justice: An EQUITAS Framework* (ED-447 Capstone, 2025).
- *Predicting Hospital Readmission Rates: A Multi-Method ML Analysis* (DSCI-633 Project Report, 2025) — ensemble and regularized regression methods for clinical risk stratification.
- *RNA Secondary Structure Prediction and Visualization: Implementing the Nussinov Algorithm with MFE Enhancements* (BIO-630 Final Project, 2025).
- *Fairness-Aware Bandits for Clinical Decision Systems* (DSCI-601 Project Report, 2023).
- *ASL Recognition Application: Image Processing for Sign Language Translation* (MS Thesis, 2015).

## Technical Skills

**Programming:** Python, R, Java, C++, C#, MATLAB, SQL, JavaScript, Bash
**Quantum Computing:** Qiskit, Cirq (gate-based quantum circuit simulation and algorithm development)
**Compiler & Systems:** LLVM/MLIR compiler backend framework; TAFFO/Precimonious (approximate computing and precision tuning)
**Machine Learning:** PyTorch, TensorFlow, scikit-learn, XGBoost, fairness-aware ML, contextual bandit workflows
**Data & Analysis:** pandas, NumPy, SciPy, SHAP analysis, statistical testing, reproducible benchmarking
**Infrastructure:** AWS, GCP, Docker, Git/GitHub, Linux, LaTeX

## Professional Experience

**VEDADATA Inc.** — Sep 2022 – Aug 2024
*Data Solutions Engineer* — *New York, NY*

- Designed and implemented data pipelines and analytics workflows using Python and cloud services.

**VIOME Inc.** — Jan 2021 – Sep 2022
*AI Data Solutions Engineer* — *New York, NY*

- Built ML workflows for healthcare applications with production deployment and monitoring support.

## Selected Fit for NTNU Quantum Compiler Technologies Position

- Directly relevant research in quantum circuit optimization and fault-tolerant routing; active study of compiler-assisted precision tuning methodologies (TAFFO, Precimonious) and their extension to QEC code selection.
- Hands-on experience with LLVM/MLIR IR-level analysis and quantum circuit frameworks (Qiskit, Cirq) providing a practical foundation for compiler backend contributions within the SPQR project.
- Reproducibility-first experimental discipline (multi-environment: local, Colab, GCP) with traceable infrastructure—well matched to SSG's open-source and benchmark-driven research culture (CGO, PLDI).
- Interest in PoliMi collaboration; familiar with TAFFO's Value Range Analysis approach as the classical precedent the SPQR project extends to the quantum domain.